



UTP
UNIVERSITI TEKNOLOGI PETRONAS

FROM RESEARCH QUESTION TO PUBLICATION

A PRACTICAL GUIDE FOR STUDENTS AND EARLY-CAREER RESEARCHERS



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Education

- Ph.D. in Information Technology**
Universiti Teknologi PETRONAS (UTP), Malaysia
- M.Sc. in Computer Science**
COMSATS University Islamabad, Pakistan
- B.Sc. in Computer Science**
University of Wah, Pakistan



Research Interests

- Artificial Intelligence and Machine Learning
- Medical and Radiological Image Analysis
- Explainable AI (XAI)
- Metaheuristic Algorithms
- Computer Vision
- Hyperspectral Imaging
- Deep Learning and Optimization
- Multimodal Data Fusion



22+
Peer-reviewed
Publications



9
h-index



8
Master's Supervised



17
Bachelor's Supervised



3
Interns Supervised



Professional Experience

- Graduate Research Fellow**
Universiti Teknologi PETRONAS (UTP), Malaysia (Current)
- University Lecturer**
University of Wah, Pakistan
- Machine Learning Engineer**
GRC Learning, Islamabad, Pakistan



Research Contributions

- Authored and co-authored 22+ publications in reputable international journals and conferences
- Research supervision and mentorship
- Peer reviewer for international journals (e.g., IEEE Access, PLOS ONE, Scientific Reports, IETE Journal of Research, EAIA, AIM)
- Active in interdisciplinary collaborations



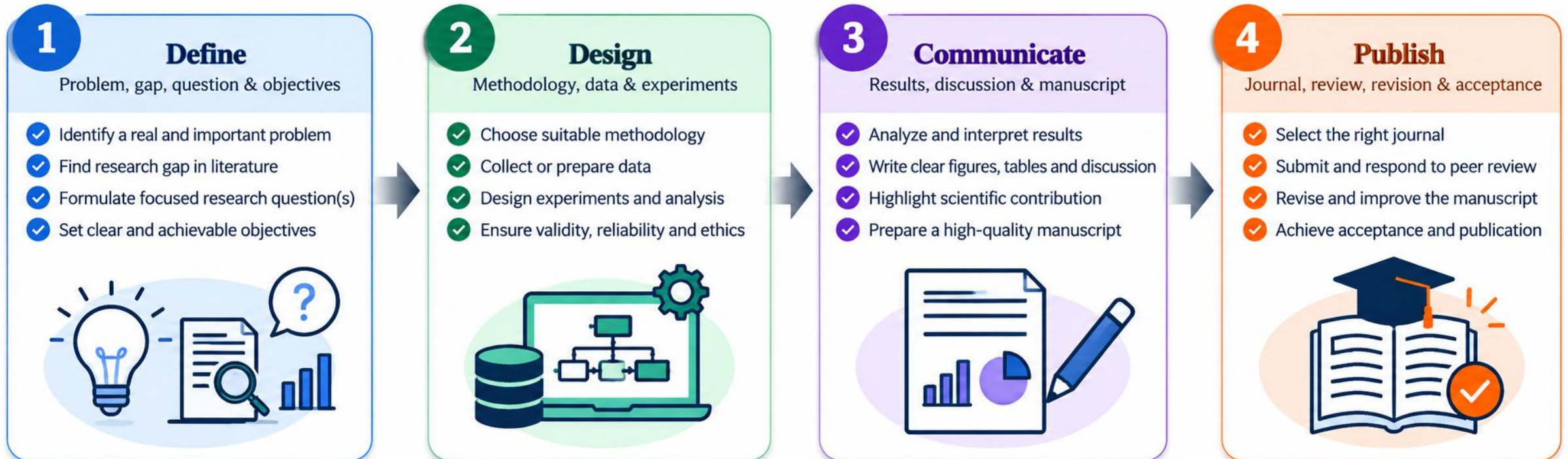
Collaborations

- Malaysia**
UTP (CeRDaS, CISIR, C2PS)
- Pakistan**
COMSATS, HITEC, University of Wah, NUCES (Karachi)
- Saudi Arabia**
Prince Sultan University (Riyadh)
- United States**
University of Texas MD Anderson Cancer Center
- Australia**
Australian universities
- Italy**
Italian universities
- France**
French universities
- Ireland**
Irish universities

OVERVIEW

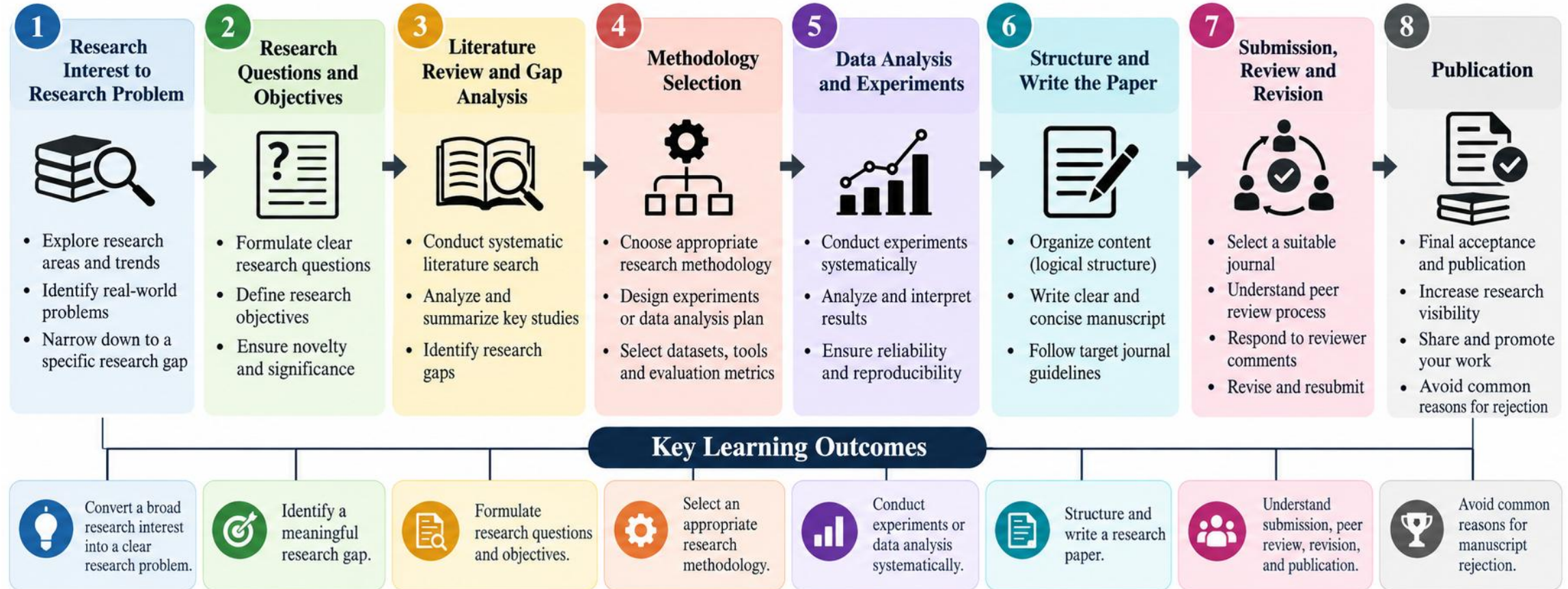
What You Will Learn

The complete path from an initial idea to a published research article



From Research Question to Publication

A Practical Roadmap for Conducting High-Quality Research and Publishing in Academic Journals



Step 1 — Selecting a Research Area

Understand the difference between: Area → Topic → Problem → Question

Area



Artificial Intelligence

A broad domain or discipline.

Field



Medical Image Analysis

A sub-area within the broader domain.

Topic



Deep Learning for Chest X-ray Classification

A specific research focus within the field.

Problem



Existing models achieve good classification performance but may provide **unreliable explanations**.

A specific issue or gap that needs to be addressed.

Research Question



How can an explainable deep-learning framework improve both classification performance and interpretability for chest X-ray analysis?

A clear and focused question to be investigated.

A Good Research Topic Should Satisfy Four Conditions



Novelty

- Is the problem new or insufficiently solved?
- Does it offer a new perspective, method, or application?



Importance

- Does solving it matter?
- Can it make a real impact (e.g., clinical, industrial, or scientific)?



Feasibility

- Can I realistically investigate it?
- Do I have the required skills, resources, and time?



Evidence

- Is sufficient data available?
- Is relevant literature available?
- Are tools and benchmarks accessible?

Step 2 — Literature Search

Before developing a research question, understand what has already been done.

1

Major Academic Databases

Scopus Scopus



ScienceDirect



Web of Science



IEEE Xplore



PubMed



SpringerLink



Google Scholar

2

Keyword Construction

Break your topic into key concepts and create keyword groups.

Example topic:

Explainable AI for
medical image classification

3

Keyword Groups

Group A – AI



“deep learning” OR
“convolutional neural
network” OR CNN

Group B – Explainability



“explainable artificial
intelligence” OR
XAI OR “Grad-CAM”

Group C – Application



“medical imaging” OR
“radiological imaging”
OR “chest X-ray”

4

Boolean Operators

AND

- Narrows results
- Retrieves records containing all concepts

OR

- Expands synonyms
- Retrieves records containing any of the listed terms

NOT

- Excludes unwanted concepts
- Removes irrelevant records

5

Combine the Keyword Groups

Use AND to combine different concepts and OR within each group.

(Group A) AND (Group B) AND (Group C)

6

Final Search Query (Example)

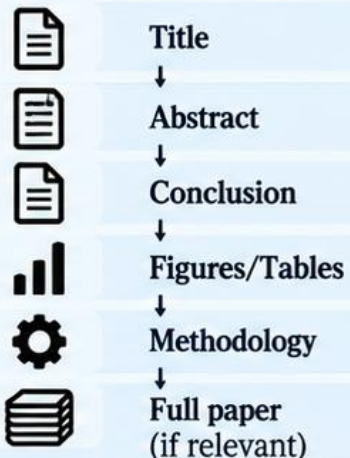
(“deep learning” OR “convolutional
neural network” OR CNN) AND
(“explainable artificial intelligence”
OR XAI OR “Grad-CAM”) AND
(“medical imaging” OR “radiological
imaging” OR “chest X-ray”)

Step 2 (Part B) – Read Literature Strategically

You do not need to read every paper from the first word to the last word.

1 Initial Screening

Scan the paper in a specific order before deciding to read the full paper.



2 Decide Relevance

After initial screening, decide whether the paper is:

- Highly relevant**
Read in detail and take notes.
- Partially relevant**
Read specific sections (e.g., method or results).
- Not relevant**
Exclude from further reading.

3 Read and Take Notes

For each important article, record key information.



- Use a spreadsheet or reference management tool (e.g., Zotero, Mendeley).
- Summarize in your own words.
- Note key findings, strengths, weaknesses, and relevance to your work.

4 Identify the Research Gap

Compare multiple studies to find:

- What has been solved?
- What are the limitations?
- What remains unexplored?
- This naturally leads to the **research gap**.

5 Information to Record from Each Important Article

Information	What to Record	Example
Problem	What problem was investigated?	Lack of explainability in medical image classification.
Dataset	What data were used?	Chest X-ray dataset (e.g., COVIDx, NIH, etc.).
Method	What approach was proposed?	Proposed CNN with Grad-CAM.
Results	What was achieved?	Accuracy, precision, recall, localization performance.
Limitation	What remains unresolved?	Limited generalization, unstable explanations, small dataset.
Future work	What do authors recommend?	Larger datasets, improved XAI methods, multi-modal analysis.

6 Key Takeaway

- Read strategically, not every word.
- Use a structured approach for screening.
- Record key information for each important article.
- Compare studies to identify limitations.
- This leads to the **research gap** for your work.

Step 3 — Understanding the Research Gap

A research gap is not simply something that has never been done. It is an unresolved limitation, unanswered question, underexplored relationship, methodological weakness, or missing evidence in existing knowledge.

1

What is Research Gap?



A research gap is an unresolved limitation, unanswered question, underexplored relationship, methodological weakness, or missing evidence in existing knowledge.

2

Why is it Important?



- Helps justify your research.
- Shows the novelty and significance of your work.
- Guides the formulation of research questions and objectives.
- Ensures your research contributes to existing knowledge.

3

Common Types of Research Gaps

Research gaps can appear in different forms:

1. Methodological Gap



Existing methods have limitations.

Example:

Existing CNN approaches achieve strong classification performance but require substantial computational resources.

2. Performance Gap



Current solutions do not achieve sufficient performance.

Example:

Existing models show good accuracy but still struggle with generalization across different datasets.

3. Knowledge Gap



An important question remains unanswered.

Example:

The impact of explainability on clinical decision-making in medical imaging remains underexplored.

4. Dataset Gap



Existing studies depend heavily on limited or nonrepresentative datasets.

Example:

Most studies use small, single-center datasets, which limits the generalizability of the results.

5. Evaluation Gap



Models are evaluated using insufficient metrics or weak validation.

Example:

Many studies report only accuracy without using robust validation or explainability metrics.

6. Interpretability Gap



Models provide predictions without adequate explanations.

Example:

High classification accuracy is achieved, but the model decisions are not sufficiently interpretable for clinical use.

7. Comparative Gap



Existing methods have not been comprehensively compared.

Example:

Limited comparison with recent deep learning and transformer-based models.

8. Application Gap



A method has worked in one domain but has not been sufficiently investigated in another.

Example:

A model performs well for chest X-ray analysis but has not been explored for CT or MRI imaging.

Step 3 (Part B) — Weak Gap vs Strong Gap

Not all research gaps are equally valuable.

1

Weak Gap



- The research problem is too general or already well addressed.
- Often results in incremental or non-novel work.
- Limited research impact and contribution.

2

Strong Gap



- The problem is specific, meaningful, and not fully solved.
- Addresses important limitations or unexplored aspects.
- Leads to novel contributions and higher research impact.

3

Comparison

Aspect	Weak Gap	Strong Gap
Scope	Too broad or general	Specific and focused
Novelty	Already well studied	Not fully explored
Impact	Incremental contribution	Significant contribution
Research value	Low to moderate	High
Example	Improving existing model with higher accuracy	Addressing an unexplored limitation or new application

4

Example: Weak Gap



Why it is weak?

- Already extensively studied.
- Only aims to achieve slightly better accuracy.
- Does not address a new problem or significant limitation.

Example Topic

Improving the classification accuracy of existing CNN models for chest X-ray images using a different optimization algorithm.

5

Example: Strong Gap



Why it is strong?

- Addresses an important and specific unresolved limitation.
- Combines performance with explainability.
- Has practical relevance and research impact.

Example Topic

Developing an explainable deep-learning framework to improve both classification performance and interpretability for chest X-ray analysis in real clinical settings.

6

From Weak Gap to Strong Gap

Weak Gap (Initial Idea)

Improve the accuracy of CNN models for chest X-ray classification.

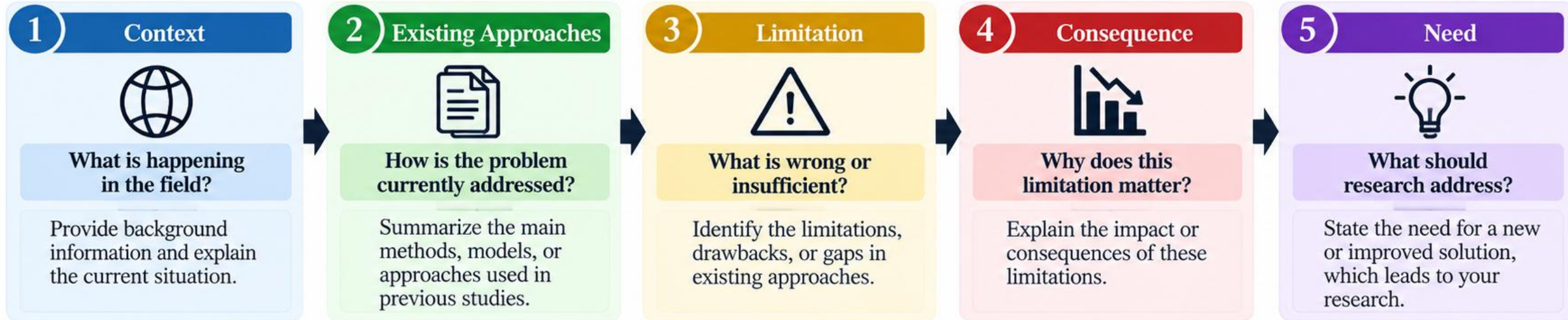


Stronger Gap (Refined Idea)

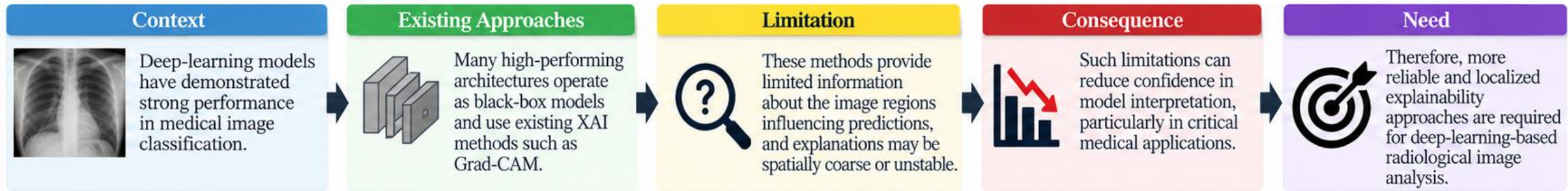
How can an explainable deep-learning framework improve both classification performance and interpretability for chest X-ray analysis, especially for real clinical applications?

Step 4 — Writing the Problem Statement

A clear and well-structured problem statement explains WHY the research is necessary.



6 Example Problem Statement



Final Problem Statement

Deep-learning models have demonstrated strong performance in medical image classification. However, many high-performing architectures operate as black-box models and provide limited information about the image regions influencing their predictions. Existing explainability methods such as Grad-CAM can generate visual explanations, but these explanations may be spatially coarse or unstable. Such limitations can reduce confidence in model interpretation, particularly in medical applications. Therefore, more reliable and localized explainability approaches are required for deep-learning-based radiological image analysis.

Step 5 — Developing the Research Question

A research question converts the identified problem into something that can actually be investigated.

1 What is a Research Question?



A research question is a clear and focused question that defines what you want to investigate.

2 Characteristics of a Good RQ

	Clear	Well-defined and unambiguous.
	Specific	Focuses on particular variables or relationships.
	Researchable	Can be investigated with available methods and data.
	Measurable	Allows empirical evaluation.
	Relevant	Addresses an important problem.
	Answerable	Can be answered within scope, time, and resources.

3 From Weak to Strong Research Question



Weak RQ (Too broad)

Is artificial intelligence useful?



Better RQ (More specific)

How can CNN based models classify chest X-ray images?



Stronger RQ (Focused and measurable)

To what extent can the proposed CNN architecture improve multi-class chest X-ray classification compared with existing deep learning methods?

4 Steps to Formulate a Research Question

- 1 Identify the research problem and research gap.
- 2 Define the key variables or concepts.
- 3 Narrow the scope to a specific aspect.
- 4 Ensure it is clear, measurable, and feasible.
- 5 Refine and validate the question.

5 Example in Medical Imaging

Weak RQ:

Is deep learning useful in medical imaging?



Better RQ:

How can CNN based models classify chest X-ray images?



Stronger RQ:

To what extent can the proposed CNN architecture improve multi-class chest X-ray classification compared with existing deep learning methods?

6 Final Tips



Align the question with your research objectives.



Use action words (e.g., how, to what extent, what factors, can, does).



Ensure data and resources are available.



Keep the question focused and realistic.



Make sure it addresses a real research gap.

1

Definition and Key Difference

Research Question (RQ)



- What do I want to know?
- States the problem in the form of a question.
- Drives the direction of the research.
- Usually one or a few questions.

Research Objective (RO)



- What will I do to find the answer?
- Describes the actions to be taken.
- Provides measurable and achievable steps.
- Usually multiple objectives.

2

Comparison

Aspect	Research Question	Research Objective
Purpose	Identifies what to investigate	Explains what to do to investigate
Form	A question	A statement (action)
Focus	Knowledge gap or problem	Specific actions and outcomes
Number	One or a few	Multiple
Example keyword	How, what, to what extent, why	Develop, design, evaluate, compare

3

Example from Medical Imaging

Research Question (RQ)



How can feature selection improve CNN-based medical image classification?



Research Objective (RO)



To develop an optimized feature-selection method for selecting discriminative CNN features and improving classification performance.

4

Good Action Verbs for Objectives

Develop	Create a new method, model or framework.
Design	Design a system or algorithm.
Evaluate	Assess the performance or effectiveness.
Compare	Compare with existing methods.
Investigate	Study the impact or relationship.
Determine	Find or identify the best approach.
Assess	Measure or analyze performance.
Validate	Test the method on different datasets.

5

Weak vs Strong Objectives



Weak Objectives (Avoid)

Understand AI in medical imaging.

Know feature selection techniques.

Learn about CNN models.

Study XAI methods.

Explore medical image datasets.



Strong Objectives (Use)

To develop a novel feature-selection method for CNN-based medical image classification.

To design an explainable deep learning framework.

To evaluate the proposed method using benchmark datasets.

To compare the proposed approach with existing state-of-the-art methods.

To validate the method for different medical imaging modalities (e.g., X-ray, CT, MRI).

6

Tips for Alignment

- 1 The research question and objectives must be consistent and aligned.
- 2 Objectives should directly answer the research question.
- 3 Use measurable action verbs.
- 4 Make objectives specific and realistic.
- 5 Ensure objectives cover key aspects (e.g., method, evaluation, comparison).
- 6 Avoid vague objectives (e.g., understand, know, learn about) unless appropriate for the research paradigm.

Step 6 — Hypotheses

A hypothesis is a testable statement about the expected relationship or difference between variables.

1

What is a Hypothesis?

- A hypothesis is a testable statement about the expected relationship, difference, or effect between variables.
- It is used mainly in quantitative research.
- Not all studies require a formal hypothesis (e.g., exploratory, qualitative, or some design studies).

A good hypothesis is clear, specific, testable, and consistent with the research question and objectives.

2

Types of Hypotheses

Null Hypothesis (H_0)

- States that there is no significant difference, relationship, or effect.
- Assumes no effect or no change.
- Used for statistical testing.

Example:

H_0 : There is no statistically significant difference between Method A and Method B in classification performance.

Alternative Hypothesis (H_1 or H_a)

- States that there is a significant difference, relationship, or effect.
- Contradicts the null hypothesis.
- Represents the expected outcome.

Example:

H_1 : Method A significantly improves classification performance compared with Method B.

3

Formulating Hypotheses

Step 1

Identify the key variables from your research question.

Step 2

Decide the expected relationship, difference, or effect.

Step 3

Write the null hypothesis (H_0).

Step 4

Write the alternative hypothesis (H_1 or H_a).

Step 5

Ensure the hypotheses are measurable and testable using appropriate statistical methods.

4

Example in Medical Imaging

Research Question



How can entropy-guided feature selection improve classification performance in deep features extracted from radiological images?



Hypotheses



H_0 : There is no statistically significant difference in classification performance between the baseline feature selection method and the proposed entropy-guided method.

H_1 : The proposed entropy-guided feature selection method significantly improves classification performance compared with the baseline method.

5

Connection with the Research Process

Research Question

Identify what you want to investigate.



Hypotheses

State expected relationship or difference.



Experiment

Conduct experiments and collect data.



Statistical Analysis

Test the hypotheses (e.g., t-test, ANOVA).



Conclusion

Accept or reject the hypotheses and answer the research question.

6

Key Points

1

Not every study requires a formal hypothesis.

2

Hypotheses are mainly used in quantitative research.

3

H_0 states no significant difference or effect.

4

H_1 states a significant difference or effect.

5

Hypotheses should be aligned with the research question and objectives.

6

Use appropriate statistical tests based on your data and experimental design.

Step 7 — Choosing the Research Methodology

The methodology explains HOW you will answer the research question.

1 Types of Research Approaches



Qualitative Research

Explores phenomena and provides in-depth understanding.

Methods: interviews, case studies, thematic analysis.



Quantitative Research

Uses numerical data and statistical analysis to test hypotheses.

Methods: experiments, surveys, statistical analysis.



Mixed-Methods Research

Combines qualitative and quantitative methods.

Methods: sequential or concurrent design.

2 Key Considerations



Research objectives

What do you want to achieve?



Nature of data

What type of data will be collected?



Available resources

Time, budget, and access to data.



Research context

Laboratory, clinical, industrial, or real-world.



Feasibility

Is the method practical and achievable?

3 Example in Medical Imaging



Research Question

How can entropy-guided feature selection improve classification performance in radiological images?



Methodology

Deep learning (CNN) for feature extraction, entropy-guided feature selection, and SVM/KNN for classification.



Data

Chest X-ray, CT, and MRI datasets.



Evaluation

Accuracy, precision, recall, F1-score, and XAI metrics (insertion/deletion, localization).



Justification

Selected due to capability to learn complex features, interpretability, and suitability for medical image analysis.

Step 7 (Part B) — Methodology Must Come from the Research Question

The methodology, data, and analysis should be directly aligned with the research question and objectives.

1 Ask These Questions

- 1 Does my methodology answer the research question?
- 2 Is the chosen method appropriate for my objectives?
- 3 Are the data and resources available?
- 4 Is the method valid and reliable?
- 5 Can the method measure the variables I am interested in?
- 6 Does the method allow me to evaluate the outcomes?
- 7 Are there limitations in the chosen method?
- 8 Do I need to use multiple methods (e.g., mixed-methods)?

2 From Research Question to Methodology (Example)

Research Question



How can entropy-guided feature selection improve classification performance in radiological images?



Research Objectives



- Develop CNN architectures.
- Design feature selection method.
- Evaluate performance.
- Provide explainability.



Methodology



- Deep learning (CNN).
- Entropy-guided feature selection.
- SVM/KNN classification.
- Focus-GradCAM for explainability.



Data



- Chest X-ray (7,135 images).
- CT (2,481 images).
- MRI (708 images).



Evaluation



- Accuracy, precision, recall, F1-score.
- Localization, fidelity, stability.
- Statistical analysis (e.g., t-test, ANOVA).



The methodology must be consistent and directly answer the research question and objectives.

Step 8 — Data Collection

Data collection provides the raw information needed to answer the research question.

1 Where does the data come from?



Public datasets

e.g., Kaggle, PhysioNet, TCIA, NIH, dataset websites



Institutional/clinical data

e.g., hospitals, research centres (with ethical approval)



Self-collected data

e.g., own experiments, imaging devices



Literature-provided data

e.g., datasets used in previous studies

2 How much data are required?



Depends on the task

e.g., classification, segmentation, detection



Consider data complexity

More classes and higher resolution require more data.



Practical guidelines

Use available datasets, data augmentation, or cross-validation when data is limited.



Quality over quantity

Diverse, representative, and well-annotated data is more important than only large size.

3 How are the data selected?



Define inclusion criteria

e.g., image modality, disease type, image quality, annotated data



Define exclusion criteria

e.g., corrupted images, low-quality scans, missing labels



Sampling strategy

e.g., random sampling, stratified sampling (class-wise)



Label verification

Ensure correct and consistent annotations from experts.

4 Is the data collection ethical and reproducible?



Ethical approval

Required for clinical or patient-based data.



Privacy and anonymization

Remove personal identifiers (e.g., name, ID, face).



Data usage permission

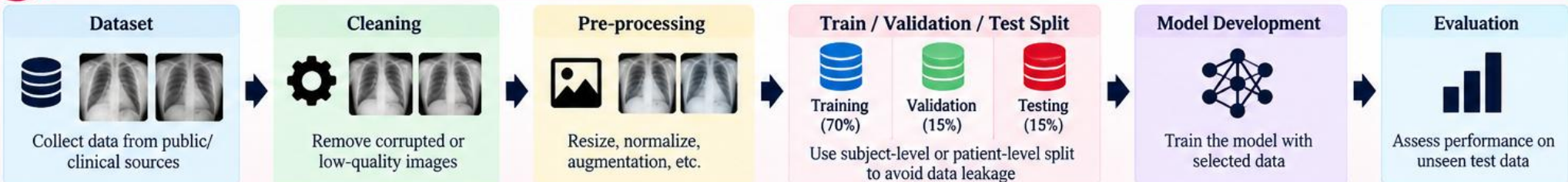
Follow dataset licenses and institutional guidelines.



Reproducibility

Provide dataset details, preprocessing steps, and experiment settings.

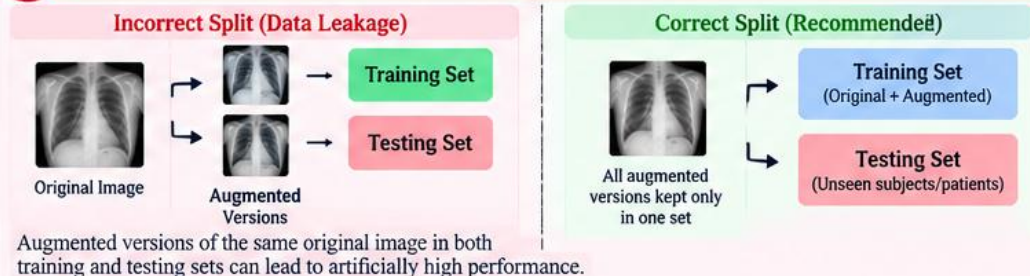
5 Data Collection Pipeline for Experimental AI Research



6 Important Considerations

- ✓ Test data should represent unseen data (not used in training or model selection).
- ✓ Avoid data leakage (e.g., same patient or augmented versions in both train and test sets).
- ✓ Ensure class balance or use appropriate strategies for imbalanced data.
- ✓ Document all data sources, preprocessing steps, and split ratios for reproducibility.
- ✓ Follow ethical guidelines and obtain necessary approvals for clinical data.

Common Data Leakage Example



Step 9 — Experimental Design

Experimental design explains how experiments are conducted to answer the research question.

1 Key Elements



Baseline models

Which existing methods will be compared against?



Evaluation metrics

What metrics will be used to measure performance?



Ablation study

Which components contribute to the improvement?



Statistical analysis

Is the improvement statistically significant?



Reproducibility

Can another researcher repeat the experiments?

2 Experimental Design Workflow



Select datasets

Define data sources and preprocessing.



Implement methods

Proposed model and baseline models.



Design experiments

Set training protocol, hyperparameters, and evaluation.



Compare and analyze

Evaluate using standard metrics and statistical tests.



Report results

Present results, ablation study, and discussion.

3 Evaluation Metrics

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Proportion of correctly classified samples.

Precision

$$Precision = \frac{TP}{TP + FP}$$

Proportion of true positives among predicted positives.

Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

Proportion of true positives among actual positives.

F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Harmonic mean of precision and recall.

4 Other Important Metrics



AUC

Area under the ROC curve (overall performance).



Sensitivity

True positive rate (same as recall).



Specificity

True negative rate.



Confusion matrix

Detailed class-wise performance.



Statistical significance

e.g., t-test, ANOVA, p-value.



Computational complexity

Time complexity and model size.



Training/Inference time

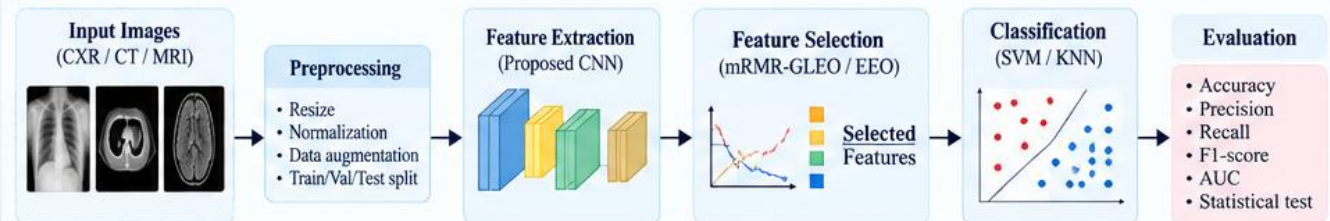
Required time for training and testing.

5 Ablation Study

Evaluate the contribution of each component by removing or replacing it.

Model / Components	Accuracy (%)	F1-score (%)
Proposed model (full)	99.98	99.98
Without feature selection	99.45	99.43
Without Focus-GradCAM	99.37	99.35
Without entropy coding	99.21	99.18
Baseline model (e.g., ResNet/EfficientNet)	98.76	98.72

6 Example Experimental Setup (Medical Imaging)



Use standard datasets, same training protocol, and appropriate statistical analysis to ensure fair comparison and reproducibility.

Step 10 — Results Are Not Discussion

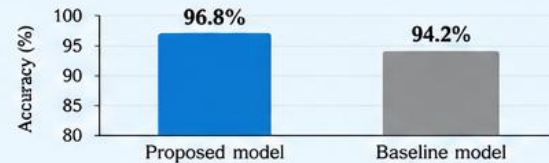
Results state what you found. Discussion explains why it happened and what it means.

1 Results: What Did You Find?

- Present objective findings without interpretation.
- Report quantitative and/or qualitative results (e.g., tables, figures, numbers).
- Use clear and concise statements.

Example (Result)

The proposed model achieved an accuracy of 96.8%, compared with 94.2% for the baseline.



2 Discussion: Why Does It Matter?

- Explain why the results occurred.
- Interpret the meaning and significance.
- Compare with existing literature.
- Discuss implications and limitations.

Example (Discussion)

The improvement may result from the proposed feature-selection mechanism, which removes redundant representations while preserving discriminative information. This is consistent with previous studies showing that feature selection can enhance model generalization in medical image classification.

3 What a Strong Discussion Should Address



Finding

Summarize the key results.



Explanation

Why did this happen? (e.g., model design, data characteristics)



Comparison with literature

How do the results compare with previous studies?



Implication

What are the practical or theoretical implications?



Limitation

What are the limitations and future work?

Step 10 (Part B) — Don't Only Report Your Best Result

Researchers expect complete and transparent evidence, not just the best-performing result.

1 Report Comprehensive Results

- Include results for all important experiments, not just the best result.
- Use the same dataset and train/test protocol across all comparisons.
- Report confidence intervals and statistical significance.
- Show results for baseline methods.
- Include ablation studies.
- Analyse and discuss any failure cases.

2 Example Results Table (Complete Reporting)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Proposed model (full)	96.8 ± 0.6	96.2	97.1	96.6
Without feature selection	95.1 ± 0.7	94.5	95.8	95.1
Without attention module	94.6 ± 0.8	93.8	95.2	94.5
Without data augmentation	94.2 ± 0.9	93.1	95.0	94.0
Baseline (ResNet)	94.2 ± 0.8	93.7	94.8	94.2
Baseline (EfficientNet)	93.6 ± 0.9	93.0	94.1	93.5

3 Key Questions Reviewers Expect Answers To

- 1 Was the same dataset used?
- 2 Was the same train/test protocol used?
- 3 Were confidence intervals reported?
- 4 Was statistical significance examined (e.g., t-test, ANOVA)?
- 5 Was there data leakage?
- 6 Were appropriate baselines included?
- 7 Was an ablation study performed?
- 8 Does the improvement justify additional complexity?



High accuracy is not automatically scientific novelty. It must be supported by proper analysis, comparison, and discussion.

Step 11 — Understanding Research Contribution

A research contribution explains what this study adds to existing knowledge.

1 What is a Research Contribution?



A research contribution is the **novel and valuable addition** that your study makes to existing knowledge, theory, methodology, or practice.

2 Types of Research Contributions



Methodological contribution

A new method, architecture, framework, or algorithm.



Empirical contribution

New experimental evidence or findings.



Theoretical contribution

A new model explanation, theoretical relationship, or conceptual insight.



Dataset contribution

A new dataset or benchmark.



Evaluation contribution

A new evaluation framework or comprehensive benchmarking.



Application contribution

A demonstrated solution to an important real-world problem.

3 Example Contributions in Medical Image Analysis

Contribution Type	Example
Methodological	A novel entropy-guided feature selection method (ECO/mRMR-GLEO).
Empirical	Comprehensive experiments showing improved classification performance on CXR, CT, and MRI datasets.
Theoretical	Analysis of feature entropy for discriminative radiological features.
Dataset	Creation of a curated multi-class chest X-ray dataset with proper split.
Evaluation	Comprehensive comparison with recent CNN architectures and XAI methods.
Application	Explainable framework for reliable diagnosis and localization of pathological regions.

Step 11 (Part B) — Novelty vs Contribution

Novelty is what is new, while contribution is what the new element adds to knowledge or practice.

1 What is Novelty?

- Refers to what is new or different in your study.
- Can be a new idea, method, dataset, application, or perspective.
- Shows originality compared to existing work.

Example (Novelty)

We introduce a new attention mechanism for CNN-based medical image analysis.

2 What is Contribution?

- Explains what the new element adds to knowledge or practice.
- Shows the value and impact of the novelty.
- Can be methodological, empirical, theoretical, dataset, evaluation, or application.

Example (Contribution)

The proposed attention mechanism improves classification robustness under limited training samples and provides interpretable visual explanations.

3 Comparison of Novelty and Contribution

Aspect	Novelty	Contribution
Focus	What is new?	What does the new element add to knowledge or practice?
Purpose	Shows originality.	Shows value and impact.
Examples	New model, new algorithm, new dataset, new perspective.	Improved performance, better interpretability, new benchmark, practical solution.
In a sentence	Introduces a new idea or approach.	Demonstrates how the new idea advances the field.

4 Example: Novelty vs Contribution in Medical Imaging

Study Element	Novelty (What is new?)	Contribution (What does it add?)
CNN architecture	A novel deep CNN (FayyazNet/MEDRADNet).	Improves feature learning and classification performance.
Feature selection	Entropy-coded EO and mRMR-coded EO.	Selects discriminative features and reduces redundancy.
XAI method	Focus-GradCAM.	Provides more reliable and interpretable visual explanations with better localization.
Dataset	Curated multi-class CXR dataset.	Enables robust evaluation and benchmarking for future research.
Application	Explainable framework for radiological analysis.	Supports reliable diagnosis and potential real-world clinical usage.

Step 12 — Writing the Research Paper

The standard IMRaD structure helps present your research in a clear, logical, and comprehensive way.



Step 12 — Writing the Research Paper (IMRaD)

The IMRaD structure helps present your research in a clear and logical way.



1 A Complete Manuscript Structure

1	Title	Clear and informative title.
2	Abstract	Brief summary of the study.
3	Keywords	4–8 relevant keywords.
4	Introduction	Background, problem, objectives, and significance.
5	Related Work / Literature Review	Review recent and relevant work.
6	Methodology	Datasets, methods, experimental design, and evaluation metrics.
7	Results	Present quantitative/qualitative findings (tables and figures).
8	Discussion	Interpret the findings, compare with literature, implications.
9	Limitations	Acknowledge the study's limitations.
10	Conclusion	Summary of contributions and future work.
11	References	Cite relevant and recent studies.
12	Supplementary Material	Additional results, datasets, code, or figures (where appropriate).

Step 12 (Part B) — How to Write an Abstract

A good abstract communicates the entire paper in a concise and complete way.

1 Use a Six-Part Formula



2 Example Abstract Structure (Medical Imaging)

- Background:** Deep-learning approaches have demonstrated considerable potential for radiological image analysis; however, existing methods remain limited by poor interpretability and redundant feature representations.
- Problem:** Current CNN-based methods often rely on heuristic feature selection and lack reliable visual explanations, leading to suboptimal classification performance and limited clinical trust.
- Objective:** This study proposes a novel explainable deep learning framework with entropy-guided feature selection for robust multi-class chest X-ray classification.
- Method:** The proposed framework integrates a deep CNN architecture, entropy-coded equilibrium optimizer for feature selection, and Focus-GradCAM for visual explanation.
- Key Results:** Experimental evaluation demonstrates an accuracy of 99.98%, outperforming existing approaches, with improved localization and stability of visual explanations.
- Conclusion:** These findings indicate that the proposed framework can enhance diagnostic performance and interpretability in medical image analysis, with potential for real-world clinical applications.

3 Key Points to Remember

- ✓ The abstract is not the introduction copied into 200 words.
- ✓ It should be self-contained and understandable without reading the full paper.
- ✓ Be concise (typically 150–250 words for journals/conferences).
- ✓ Include the main problem, method, and key results.
- ✓ Use clear and direct language.

Step 12 (Part C) — Choosing a Good Paper Title

A good title attracts readers and clearly communicates the main contribution.

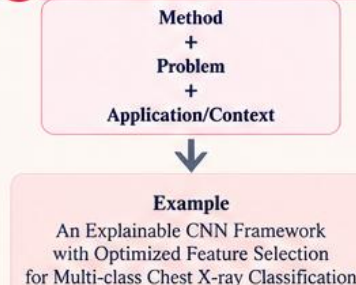
1 Examples of Paper Titles

	Weak (Too general) Deep Learning for Medical Images	• Too broad and unclear • Does not specify the task, method, or contribution • Not informative
	Better (More specific) Explainable Deep Learning for Multi-class Chest X-ray Classification	• Specifies the domain and task • Indicates the use of explainability • Still missing details about the method or novelty
	Stronger (Clear and informative) An Explainable CNN Framework with Optimized Feature Selection for Multi-class Chest X-ray Classification	• Clearly indicates the method (CNN) • Mentions a key contribution (optimized feature selection) • Specifies the application (multi-class chest X-ray classification)

2 A Good Title Should Include

- Method**
What approach or technique is used? (e.g., CNN, Transformer, optimization, XAI)
- Problem**
What problem or task is addressed? (e.g., disease classification, segmentation)
- Application/Context**
Where is it applied? (e.g., chest X-ray, MRI, hyperspectral imaging)

3 Title Formula



4 Avoid Unnecessary Phrases

- Avoid**
 - A New
 - A Novel
 - An Innovative
 - A Hybrid
 - A Combined
 - A Study on
 - A Review of
 - A Comprehensive
 - State-of-the-Art
- Instead**
 - Use clear and direct wording.
 - Demonstrate novelty through the actual method, results, or problem.
 - Be specific and informative.

Step 13 — Figures and Tables

Well-designed figures and tables help readers understand the main findings quickly and clearly.

1 Why Figures and Tables Are Important?

- Summarize complex information.
- Make the paper easier to read and understand.
- Highlight key findings.
- Allow quick comparison of methods and results.
- Support your research questions.
- Should be self-contained with clear captions.



2 Common Types of Figures



Architecture

Model or system architecture.



Framework

Overall framework or pipeline.



Workflow

Step-by-step process.



Training curve/Trend

Model performance over epochs.



Visual comparison

Performance comparison (bar/line charts).



Experimental results

Qualitative examples (e.g., sample images, heatmaps).

3 Common Types of Tables



Numerical comparisons

Performance of different methods (accuracy, precision, etc.).



Dataset information

Dataset sources, size, classes, and splits.



Literature comparisons

Comparison with existing studies.



Ablation results

Effect of each component.

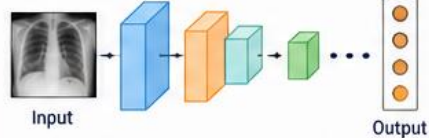


Hyperparameters

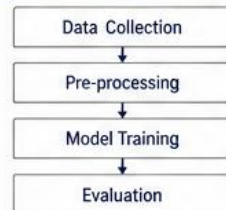
Training settings and parameter values.

4 Examples of Figures

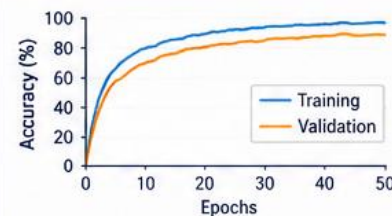
(a) Architecture



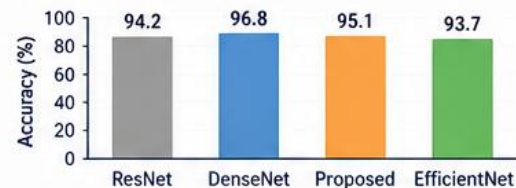
(b) Framework / Workflow



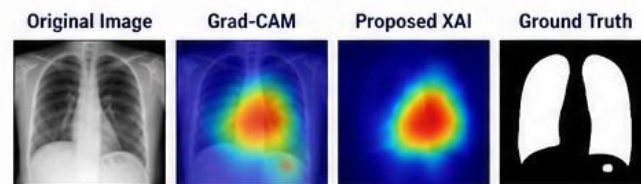
(c) Training Curve



(d) Visual Comparison



(e) Experimental Results (Qualitative)



5 Example of Tables

(a) Performance Comparison Table (Numerical)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
ResNet	94.2	93.8	94.1	93.9
DenseNet	95.1	94.6	95.0	94.8
EfficientNet	93.7	93.1	93.5	93.3
Proposed	96.8	96.2	97.1	96.6

(b) Dataset Information Table

Dataset	Modality	Classes	Total Images	Train	Test
Chest X-ray	CXR	4	7,135	4,997	2,138
COVID-19 CT	CT	2	2,481	1,738	743
Brain Tumor MRI	MRI	2	706	496	212



Every figure and table should answer a research-relevant question, have a clear caption, and be properly referenced in the text.

Step 14 — References and Citation Management

Proper referencing gives credit to original authors, improves the credibility of your work, and helps readers find related sources.

1 What is Reference Management?

- Organizing and managing your citations using software tools.
- Helps you insert, format, and update citations automatically.
- Saves time and reduces errors.
- Ensures consistent citation style throughout the paper.



2 Common Reference Managers



Zotero

- Free and open source
- Easy to use, good for beginners



Mendeley

- Reference management and PDF tools
- Good for collaboration



EndNote

- Powerful features
- Popular in research institutions

3 Good Citation Practice



Cite original methods

Give credit to the original authors of methods, models, or algorithms.



Cite datasets

Provide proper citations for publicly available datasets.



Cite important theoretical claims

Support key statements with relevant literature.



Cite previous findings

Compare and discuss your work with existing studies.



Cite evaluation methods

Reference standard evaluation metrics and protocols.

4 What to Cite in a Research Paper?



Original methods

Algorithms, models, and techniques.



Datasets

Public datasets used in the study.



Important theoretical claims

Key concepts, definitions, or frameworks.



Previous findings

Related work and similar studies.



Evaluation methods

Metrics, protocols, and baseline models.

5 In-text Citation Examples

Single author

Deep learning has shown promising results in medical imaging (He, 2016).

Two authors

Recent studies have investigated this approach (Chen and Wang, 2020).

Multiple authors

This method has been widely used in radiology (Khan *et al.*, 2021).

6 Reference List Example (APA Style)

- [1] He, K. (2016). Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [2] Chen, L., & Wang, Y. (2020). Attention-based CNN for medical image classification. *Computers in Biology and Medicine*, 123, 103912.
- [3] Khan, A., Ali, S., & Rahman, M. (2021). Multi-class chest X-ray classification using deep learning. *Journal of Medical Imaging*, 8(2), 1–12.
- [4] World Health Organization. (2020). COVID-19 imaging dataset. <https://doi.org/10.5281/zenodo.3757476>



Always cite the original source when available and keep a consistent citation style throughout your paper.

Step 15 — Selecting the Right Journal

Choose a journal that fits your research scope, audience, and article type.

1 Consider Journal Fit Early



- Check if the journal publishes your type of research.
- Read recent articles to understand the scope and style.
- Do not complete the manuscript and then search for a journal.

2 Evaluate the Journal



- Aims and scope
- Type of articles published
- Quality and reputation
- Editorial board
- Average review time
- Acceptance rate
- Publication model (subscription, hybrid, or open access)
- Publication fees (APC)

3 Audience



- Who should read your paper?
- Researchers, clinicians, practitioners, or a broader audience?
- Choose a journal that reaches your target audience.

4 Indexing



- Check if the journal is indexed in Scopus, Web of Science, PubMed, IEEE Xplore, or other relevant databases.
- Indexing improves visibility and citation potential.

5 Quality Indicators



- Use indicators such as:
 - CiteScore (Scopus)
 - Journal Impact Factor (JCR)
 - SJR (Scimago)
 - h-index
- Consider these together rather than relying on a single metric.

6 Article Type



- Ensure the journal accepts your manuscript type:
 - Original research
 - Review article
 - Short communication
 - Case study
 - Technical note, etc.

7 Publication Model



- Subscription (no fee for authors)
- Hybrid (optional open access)
- Open access (usually requires APC)
- Check funding availability for APC if needed.

8 Scope



- Read the aims and scope carefully.
- Check if your topic fits (e.g., AI, medical imaging, explainability).
- Cook at recently published papers with similar topics.

9 Review/Publication Timeline



- Check the average review time.
- Consider the overall timeline to publication (review, revision, acceptance, production).
- Ensure it meets your requirements (e.g., graduation, project deadline).

10 Additional Considerations



- Check for predatory journals (avoid).
- Read author guidelines carefully.
- Look at the editorial board.
- Review recently published articles.
- Consider journal ranking and reputation.
- Ensure ethical and transparent publication practices.
- Check copyright and data sharing policies.
- Consider potential audience and impact.



Choose a journal based on relevance, quality, and audience, not only on a high impact factor.

Step 15 (Part B) — Journal Scope vs Impact Factor

Journal scope is more important than impact factor alone.

1 Why Scope Matters More?



- A high-impact journal may not be the right fit if your topic is outside its scope.
- A well-matched journal is more likely to accept your paper, even if its impact factor is lower.
- Journal selection is a matching problem, not simply a ranking problem.

2

✗ Wrong Approach (Impact Factor Only)

- Choosing a journal only because it has a high impact factor.
- Ignoring the journal's aims and scope.
- Higher chance of rejection.
- Wastes time and delays publication.

✓ Right Approach (Scope + Quality)

- Choose a journal that matches your topic and research goals.
- Ensure the scope covers your research area (e.g., AI, medical imaging, explainability).
- Then consider quality indicators (impact factor, CiteScore, SJR, etc.).
- Check audience, article type, and publication model.

3 Example

Your paper is about:
AI + Medical Imaging + Explainability

Journal	Scope Match	Impact Factor (CRF/JCR)	Decision
Nature Medicine	Low (mainly clinical medicine)	High (50+)	✗ Not suitable
IEEE Transactions on Medical Imaging	High (medical imaging, AI)	High (10+)	✓ Suitable
Computers in Biology and Medicine	High (AI in healthcare)	Medium (7+)	✓ Suitable
Expert Systems with Applications	High (AI applications)	Medium (6+)	✓ Suitable

4 Key Takeaways



- Journal scope determines the relevance and suitability of your work.
- Impact factor alone does not guarantee acceptance.
- A lower impact factor journal with the right scope can have higher value for your research.
- Always read the aims and scope, and recent articles, before submission.
- Selecting the right journal increases the chances of acceptance and ensures your work reaches the right audience.

Step 15 (Part C) — Avoid Predatory Journals

Avoid predatory journals to protect the quality, credibility, and impact of your research.

1 What are Predatory Journals?



- Predatory journals are fake or low-quality journals that prioritize profit over scholarly publishing.
- They often lack proper peer review, editorial standards, and transparency.
- Publishing in such journals can harm your reputation and research career.

2 Warning Signs



Unsolicited email invitations.



Promises of very fast publication (e.g., few days).



Fake or misleading impact factors.



Unclear or fake editorial board.



Lack of transparent peer review process.



High publication fees without clear details.



Poorly written website or scope.



Editors' names are not verifiable.

3 How to Check a Journal?



Check if the journal is indexed in trusted databases (Scopus, Web of Science, PubMed, DOAJ).



Verify the publisher (e.g., IEEE, Elsevier, Springer).



Look at the editorial board (real and verifiable experts).



Read recent published articles (quality and relevance).



Check the journal's aims and scope.



Look for a clear peer review policy.



Compare publication fees with genuine journals.



Check for inclusion in Beall's list or other predatory journal checkers (e.g., Cabells, Think.Check.Submit).

4 Consequences of Publishing in Predatory Journals



Your work may not be properly reviewed.



Low visibility and no real impact.



Can damage your academic reputation.



May not be indexed in major databases.



Can affect future funding, job, or PhD applications.

5 Useful Tools and Resources



Think. Check. Submit.

<https://thinkchecksubmit.org>



Journal Checker Tool (JCT)

<https://jane.biosemantics.org>



Cabells Predatory Reports

<https://www.cabells.com>



Scopus Source List

<https://www.scopus.com/sources.uri>



Web of Science Master Journal List

<https://mjl.clarivate.com>



DOAJ (Open Access Journals)

<https://doaj.org>

6 Example: Comparison

Feature	Genuine Journal	Predatory Journal
Peer review	Rigorous and transparent	Often absent or unclear
Editorial board	Well-known experts	Fake or unverifiable
Indexing	Scopus, WoS, PubMed, etc.	Not indexed or fake claims
Publication time	Reasonable (weeks–months)	Very fast (few days)
Publication fee	Clearly stated and reasonable	High or hidden fees
Journal scope	Relevant and specific	Very broad and unrelated
Website quality	Professional and informative	Poorly written with errors
Impact factor	Real and verifiable	Fake or misleading



Key Takeaways

- Be careful and verify a journal before submission.
- Use trusted databases and journal checkers.
- Do not be attracted by fast publication or low review standards.

- Choose reputable publishers and journals that match your research scope.
- Publishing in a genuine journal increases the visibility, credibility, and impact of your work.



Step 16 — Before Submission

Use this manuscript checklist to ensure your paper is complete, high quality, and ready for submission.

1

Scientific Checklist



Research question answered

The study clearly addresses the research question(s).



Novelty clearly established

The contribution and what is new are clearly stated.



Strong baselines

Appropriate comparison methods and baselines are included.



Appropriate evaluation

Suitable metrics, statistical analysis, and validation are performed.



Limitations acknowledged

The study's limitations are clearly discussed.



2

Writing Checklist



Clear title

The title is concise, informative, and reflects the main contribution.



Structured abstract

The abstract is well-structured and easy to understand.



Logical introduction

Background, problem, objectives, and significance are clearly explained.



Reproducible methodology

Methods are described in sufficient detail for reproducibility.



Results and discussion aligned

Results are properly analyzed and discussed with literature.



Conclusion answers the question

The conclusion summarizes the findings and answers the research question(s).



3

Journal Checklist



Scope checked

The journal scope matches your topic and research goals.



Author guidelines followed

Manuscript follows the journal's author guidelines.



Template followed

Correct journal template and formatting are used.



Reference style checked

Citations and reference style follow the journal's requirements.



Figures meet requirements

All figures and tables are clear, correctly formatted, and of high quality.



Supplementary files prepared

Additional files (datasets, code, appendices) are ready (if required).



4

Final Check Before Submission



Proofread manuscript



Check all checklist items



Get feedback from co-authors/supervisors



Prepare files and submit



Tip:

Use this checklist to ensure your manuscript is complete, high quality, and ready for submission to the selected journal.

Step 17 — Cover Letter

A cover letter introduces your manuscript and explains why it is suitable for the journal.

1 What to Include?



Manuscript title

Mention the full and exact title of your paper.



Problem

Briefly state the research problem and its significance.



Main contribution

Highlight the key novelty and contributions of your work.



Why it matters

Explain the value and potential impact of your study.



Why this journal is appropriate

Show how your work fits the journal's scope and audience.



Organizational/rationale

Provide a short overview of the study and why reviewers will find it interesting.



Conclude

Politely request the editor to consider your manuscript for publication.

2 Tips for a Strong Cover Letter



Keep it concise (1 page).



Be clear and professional.



Highlight novelty and significance.



Match the journal's scope.



Avoid repeating the abstract.



Follow the journal's specific guidelines.

3 Sample Structure

Dear Editor,



Paragraph 1: Manuscript title and submission type



Paragraph 2: Research problem and significance



Paragraph 3: Main contribution and novelty



Paragraph 4: Why this journal is appropriate



Paragraph 5: Brief conclusion and polite request

Sincerely,
[Your Name]

4 Example Opening

Dear Editor,

Please find enclosed our manuscript entitled “A Novel Explainable Deep Learning Framework for Chest Disease Classification” for consideration as a research article in [Journal Name]. This study addresses the problem of accurate and interpretable medical image analysis by proposing a new deep learning framework with improved explainability and superior diagnostic performance. Given the journal's focus on medical imaging and artificial intelligence, we believe our work is well aligned with its scope and will be of interest to its readership.

Step 18 — Submission Process

Follow the typical workflow from submission to publication.

1 Manuscript Submission



- Ready all required files (manuscript, figures, tables, cover letter, supplementary materials).
- Submit through the journal's online system.

2 Technical Check



- Journal checks compliance with formatting, scope, and basic requirements.
- May be returned for minor corrections.

3 Editor Screening



- Editor assesses relevance, novelty, and quality.
- Possible outcomes: desk rejection or send for peer review.

4 Peer Review



- Assigned to 2–3 expert reviewers.
- Reviewers provide detailed comments and recommendations.

5 Revision (if required)



- Revise the manuscript based on reviewer and editor comments.
- Types: Minor Revision / Major Revision.

6 Re-submission



- Submit the revised manuscript with a response letter addressing each comment.
- Editor and/or reviewers re-evaluate the revision.

7 Acceptance



- Manuscript is accepted for publication.
- Complete any final checks (e.g., copyright form, publication charges if applicable).

8 Proof



- Check the galley proof for minor errors.
- Approve the final version.

9 Publication



- Your paper is published online and/or in an issue.
- Share and promote your published work.

Possible Editor Decisions



Accept

The paper is accepted for publication.



Minor Revision

Small changes required. Usually one round.



Major Revision

Substantial changes required. May involve multiple rounds.



Reject

The paper is not suitable for the journal.

Review Flow



! Key Takeaway

Understand the submission process and be prepared for revisions. This is a normal part of academic publishing.

Understanding Reviewer Comments

Reviewer comments are not meant to criticize you personally, but to improve the quality, clarity, and **impact** of your paper.



Reviewer's Perspective

Reviewers assess the quality, originality, and significance of the research from an expert point of view.



Why They Give Comments?

- To improve the clarity and quality of your work.
- To ensure the methodology is sound.
- To check the validity of the results and conclusions.
- To strengthen the novelty and contribution.



Before Responding

- Read all comments carefully and completely.
- Understand the main concerns.
- Keep a professional and positive attitude.
- Do not take comments personally.



Structure of a Reviewer Comment

Most comments focus on one or more of the following:



Comment

Reviewer's observation or concern



Response

Your detailed explanation or revision



Action

Changes made in the manuscript



Location of change

Specify section, figure, table, and page number

Response-to-Reviewers Strategy

A clear and respectful response helps the editor and reviewers understand how you have addressed their comments.

1

Thank the Reviewer



- Start with a polite and sincere thank you.
- Show appreciation for their valuable time and constructive feedback.

2

State the Changes



- Clearly mention what changes you have made.
- Indicate whether the comment is fully, partially, or not addressed (with justification).

3

Explain Your Reasoning



- Provide a clear and logical explanation.
- Support your response with evidence, references, or additional analysis.

4

Be Specific



- Describe exactly what was changed in the manuscript.
- Mention specific section, figure, table and page/line numbers.

5

Identify What is New



- Highlight new content, additional experiments, revised analysis, or updated discussion.
- Make it easy for the reviewer to locate the changes.

Example Response

Reviewer Comment:

The dataset description is not clear. Please provide more details about data sources and preprocessing steps.

Response:

Thank you for this valuable comment. We have expanded the dataset description in the revised manuscript to clearly explain the data sources and preprocessing steps.

Action:

Added detailed dataset information and preprocessing procedure.

Location of Change:

Section 3.1 (Dataset), Page 5, Lines 120–145.

Major Revision Is Not Rejection

A major revision means the editor sees potential in your work and expects substantial improvements before acceptance.



What It Means

- The journal considers your work worthy of publication, but requires major improvements.
- It is an opportunity to strengthen your paper.
- Many high-quality papers are accepted after major revision.



Tips for Handling Major Revision

- Address all major and minor comments carefully.
- Provide a detailed, point-by-point response.
- Make clear and visible changes in the manuscript.
- Use tracked changes if required.
- Improve figures, tables, discussion, and references as suggested.
- Maintain a professional and polite tone.

Example: Major Revision Response Table

Reviewer	Comment	Action	Updated Location	Status
R1	Add baseline comparison.	Added experiments with state-of-the-art methods.	Sec. 4.3 (p. 12)	Done
R1	Explain dataset in more detail.	Expanded dataset description and preprocessing steps.	Sec. 3.1 (p. 5)	Done
R2	Novelty is unclear.	Revised introduction to clearly highlight the novelty and contributions.	Sec. 1.2 (p. 3)	Done

Rejection: Key Points and Actions

A rejection is a decision, not the end of your research. Use the feedback to improve your work.

Rejection ≠ Research Failed



**RESEARCH
FAILED**

≠



**OPPORTUNITY
TO IMPROVE**

- Rejection is a common part of the research process.
- It often provides valuable feedback for improvement.
- Many high-quality papers are accepted after one or more rejections.

Common Reasons for Rejection



Journal mismatch

The topic is outside the journal's scope.



Novelty presentation

The contribution is not clearly established or sufficiently novel.



Methodological weakness

Methods are not robust or clearly explained.



Insufficient experiments

Lack of comparison, ablation, or validation.



Writing/presentation

Unclear writing, poor organization, or language issues.



Reviewer concerns

Questions about validity, significance, or clarity.



Editorial priorities

The journal may prioritize other topics or papers.

3 What to Do After Rejection



Read and Understand

- Carefully read all reviewer and editor comments.
- Identify the main reasons for rejection.



Diagnose

- Determine which issues are major and minor.
- Check if the journal is the right fit.



Improve

- Address all valid concerns.
- Add new experiments, analysis, or clarification.



Reposition

- Consider a more suitable journal.
- Tailor the manuscript according to the new journal's scope.



Resubmit

- Submit the revised manuscript.
- Mention previous submission if required by the journal.

Key Takeaways



- Rejection is a normal and common experience.
- Use reviewer feedback constructively.
- Address methodological, experimental, and writing issues.
- Consider a more suitable journal if the scope is not aligned.
- Do not immediately resubmit the unchanged manuscript.
- Stay motivated and keep improving your work.

Typical Workflow After Rejection



Research Ethics: Major Issues and Good Practices

Maintain research integrity for credible, responsible, and impactful research.

1 Plagiarism



Using another person's work without proper attribution.

Examples:

- Copying text, figures, or ideas without citation.
- Paraphrasing without proper reference.
- Self-plagiarism (reusing your own published work).

Good Practice:

Always cite sources and use plagiarism checking tools.

2 Self-plagiarism / Text Recycling



Reusing substantial parts of your previously published material inappropriately.

Examples:

- Copying text, results, or figures from your own papers.
- Submitting the same content to multiple journals.

Good Practice:

Clearly cite your previous work and explain the differences.

3 Data Fabrication



Inventing or creating data that did not actually exist.

Examples:

- Generating fake data or results.
- Reporting experiments that were never conducted.

Good Practice:

Report only genuine and verifiable data.

4 Data Falsification



Manipulating research data or results dishonestly.

Examples:

- Altering or selectively removing data
- Changing results to support a desired conclusion.

Good Practice:

Present data honestly and transparently.

5 Image Manipulation



Modifying scientific images in misleading or unethical ways.

Examples:

- Enhancing, deleting, or altering image features.
- Misrepresenting medical or experimental images.

Good Practice:

Apply only acceptable adjustments and follow field guidelines.

6 Authorship Problems



Adding authors without legitimate contributions or excluding legitimate contributors.

Examples:

- Gift, guest, or honorary authorship.
- Excluding contributors who made significant contributions.

Good Practice:

Follow authorship criteria (e.g., ICMJE guidelines).

7 Duplicate Submission



Submitting the same manuscript to multiple journals simultaneously.

Examples:

- Submitting to more than one journal at the same time.
- Submitting a similar manuscript with minor changes.

Good Practice:

Submit to one journal at a time and disclose related work.



Key Takeaways

- Research ethics are about honesty, transparency, and responsibility.
- Maintain integrity at every stage of your research.
- Follow institutional and journal ethical guidelines.
- Use plagiarism checking and data verification tools.
- Give proper credit to all contributors.
- Ethical research builds credibility and long-term impact.

Publication Formula

A strong publication is the result of several key elements working together.

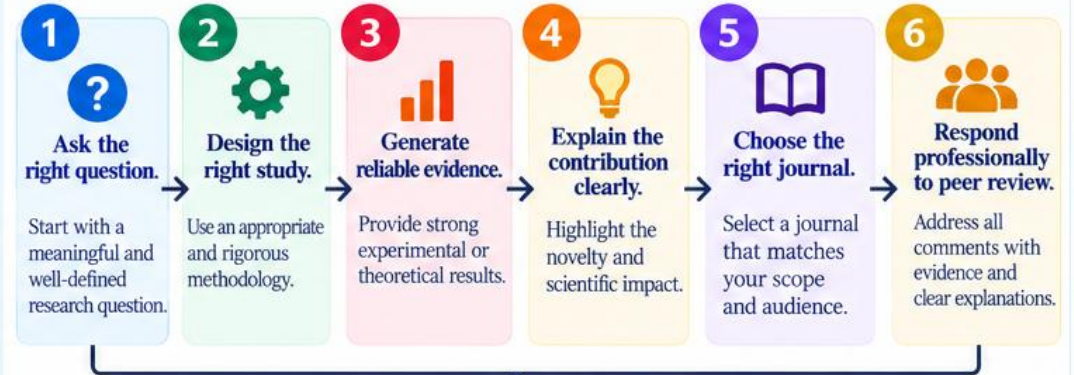
- 1 Important Problem**
Address a real and significant research problem.
- 2 Clear Research Gap**
Identify what is missing in existing literature.
- 3 Focused Research Question**
Define a specific and answerable research question.
- 4 Appropriate Methodology**
Use a suitable and well-designed methodology.
- 5 Reliable Evidence**
Provide solid experimental or theoretical evidence.
- 6 Clear Contribution**
Clearly state the novelty and contributions.
- 7 Strong Academic Writing**
Write clearly, logically, and professionally.
- 8 Right Journal**
Choose a journal that fits your topic, scope, and audience.
- 9 Effective Response to Reviewers**
Address all comments professionally with evidence.



- ✓ Accepted by a reputable journal
- ✓ Adds value to the field
- ✓ Cited and used by others
- ✓ Enhances your research profile
- ✓ Opens opportunities (funding, collaboration, career growth)

Final Takeaway

Follow this sequence to turn your research into a successful publication.



Publish.

“Do not start your research by asking, “Where can I publish?”
Start by asking, “What important question can I answer?””

Key Message

Research publication is not simply about writing a manuscript. It is a chain of logically connected decisions. A strong paper begins with a meaningful problem and a well-defined research question. The methodology must answer that question, the results must provide reliable evidence, and the discussion must explain the scientific contribution. When these elements are aligned, publication becomes the natural final stage of the research process.

Remember

- Focus on solving an important problem.
- Ensure alignment between objectives, methodology, and results.
- Maintain research integrity and high academic standards.
- Be persistent and learn from feedback.
- Publication is a milestone, not the end — it leads to new ideas and opportunities.

THANK YOU EVERYONE



for your time, attention, and valuable support
throughout this session.



QUESTION AND ANSWER

